Sounds of San Anto — Comprehensive Dataset Analysis & External Connection Mapping

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Section 1: Dataset Summary

Overview

The Sounds of San Anto dataset is a multi-sheet Excel workbook documenting concert and performance history in San Antonio, Texas, spanning **1857 to 2023** — roughly 166 years of live music. It captures venue, artist, event, date, genre, and geographic information at varying degrees of completeness across sheets.

Sheet Inventory

Sheet	Rows	Description
`Venue_Artist_Event`	37,892	**Primary event log** — full temporal range, all artists/venues
`Ven_Ev_Ar_Address`	20,638	Geocoded subset (1970–2010) with ZIP, lat/lon, address
`venues_with_coordinates`	974	Venue registry with coordinates (312 geocoded, 662 pending)
`Layer 1 Venues`	~300+	Venue list with address/ZIP
`artistsGenre`	4,797	Artist–genre mapping (many-to-one artist)
`genreIndex`	985	Unique genre lookup
`artistsIndex`	1,769	Unique artist lookup
`Sheet5`	~500+	Artist verification list with name variations
`Venues No Address`	Small	Venues lacking address data

Variable Inventory

Primary Sheet: `Venue_Artist_Event`

Variable	Type	Notes
`Month`	Categorical (string)	12 months + 1 anomalous "1" value — needs cleaning
`Day`	Numeric (string)	Day of month
`Year`	Numeric	1857–2023; pre-1940 records are sparse
`Venue`	Categorical (string)	2,603 unique venues
`Event_Artists`	Free text	Raw string, multiple artists per event
`Artist_Formula`	Categorical (parsed)	20,341 unique parsed artist names
`Event_Formulas`	Categorical	79.6% null — event names only partially populated
`Index`	Numeric (ID)	Row identifier

Geocoded Sheet: `Ven_Ev_Ar_Address`

Variable	Type	Notes
`Address`	Text	15% missing
`Zip`	Numeric	15% missing; 15 unique ZIP codes dominate
`Longitude` / `Latitude`	Numeric	68.8% missing — major gap
`Artist`	Categorical	Parsed single-artist rows
`Event`	Categorical	78.2% null
`City` / `State`	Categorical	All San Antonio / Texas where present

Key Patterns & Anomalies

Temporal distribution:

- Pre-1940: sparse (<500 events across 90 years) — collector bias toward modern era
- 1980s: peak decade (~6,766 events) — aligns with SA's population and entertainment boom
- 1990–1995: documented dip — potential data gap, economic recession (1990–91), or post-Gulf War effect
- 2020s: 6,573 events in ~3 years — likely incomplete but suggests post-COVID surge

Geographic gaps:

- Only 312 of 974 venues (~32%) are geocoded

- Most geocoded events cluster on the **north side** of SA — the journal explicitly flags this as a research question regarding southside economic impact
- Top ZIPs: 78205 (Downtown), 78212 (Midtown/Alamo Heights border), 78219 (East Side)

Genre data:

- 985 micro-genres for ~1,769 indexed artists — extreme fragmentation
- "Unknown" is the 2nd most frequent classification (137 records)
- Collector bias toward Country clearly present; Tejano likely severely undercounted
- Many artists map to multiple genre entries (one-to-many relationship)

Missing/derived:

- `Event_Formulas` is 80% null — limits event-type analysis
- No ticket price, attendance, or revenue data
- No artist origin/hometown data
- No venue capacity data
- Day-of-week is derivable (Month + Day + Year → `lubridate::wday()`)
- Season is derivable
- "Era" clustering is derivable (pre-WWII, postwar, classic rock era, etc.)

Potential Derived Variables

Derived Variable	Source Fields	Method
Full date	Month + Day + Year	`lubridate::mdy()`
Day of week	Full date	`wday()`
Season	Month	Recode
Decade	Year	`(Year %/% 10) * 10`
Macro-genre	`Genres`	Parent-category lookup table
Artist type	`Artist_Formula`	NLP or manual: solo vs. band vs. ensemble
Venue neighborhood	Lat/Lon	Spatial join to SA neighborhood shapefile
Venue Census tract	Lat/Lon or ZIP	Spatial join or ZIP-to-tract crosswalk
Southside binary	Lat/Lon	`Latitude < 29.42` (approximate TX-151 boundary)
Event gap (days)	Sorted by Venue+Date	Days between consecutive events at same venue

Derived Variable	Source Fields	Method
Artist first/last SA appearance	Artist + Year	`min()` / `max()` by artist
Venue active lifespan	Venue + Year	`max(Year) - min(Year)`

Section 2: External Dataset Connection Map

Table 1: Variable → External Dataset → Merge Key → Potential Insights → Priority

#	Primary Variable	External Dataset	Source	Merge Key	Po
1	ZIP Code	**ACS 5-Year Estimates (Census)** — B19013 Median HH Income	Census Bureau / `tidycensus`	ZIP → ZCTA crosswalk	Do coc cor mu inc
2	ZIP Code	**ACS** — B02001 Race/Ethnicity by ZIP	Census / `tidycensus`	ZIP or tract	Do cor His pop unc tes
3	Lat/Lon	**SA Neighborhood Boundaries** (City of San Antonio Open Data)	data.sanantonio.gov	Spatial join	As: gec nar ena lev
4	Lat/Lon	**Census Tract Boundaries** (TIGER/Line)	Census TIGER	Spatial join	Joi der (inc edu eve
5	ZIP / Tract	**ACS** — Educational Attainment (B15003)	Census / `tidycensus`	ZIP or tract	Is I stra nei edu

#	Primary Variable	External Dataset	Source	Merge Key	Po
6	Year	**BLS CPI / Inflation Data**	BLS / FRED	Year	No ec rea tick ad
7	Year	**Bexar County Population Estimates**	Census / Texas Demographic Center	Year	No yea gro 198 po de
8	Year	**National Bureau of Economic Research (NBER) Recession Dates**	NBER	Year range	Ov pe fre tes 95 ec
9	Genre + Year	**Billboard Chart Archives** (Whitburn project / public compilations)	Academic datasets	Genre + Year	Dic po co SA of tre
10	Venue + Address	**Bexar County Appraisal District (BCAD)** — Property Tax Records	bcad.net (public)	Address	Co lon val ide ve
11	Venue + ZIP	**SA City Business License Data**	data.sanantonio.gov	Address/ZIP	Cr act re th
12	ZIP / Tract	**ACS** — B08301 Means of Transportation to Work	Census	ZIP or tract	Pr de in t ma ac in
13	Month / Season	**NOAA Climate Data Online** — SA	NOAA	Month + Year	Do co mo

#	Primary Variable	External Dataset	Source	Merge Key	Po
		weather/temperature records			sur sup
14	Lat/Lon	**SA Urban Heat Island Data**	City of SA / NOAA	Spatial	Ov ge vul len ac
15	Artist	**MusicBrainz / Discogs Open Data**	musicbrainz.org	Artist name	Ad sta (cr
16	Artist	**Spotify API / Last.fm data** (historical)	Spotify/Last.fm	Artist name	Str pro wh pre
17	Genre (Tejano)	**Tejano Music Awards Archives**	tejanomusic.com / web scrape	Artist name	Ide Tej col
18	Venue + ZIP	**HUD Fair Market Rents**	HUD USER	ZIP	Pro nei ge on
19	ZIP / Tract	**CDC PLACES Dataset** — health outcomes by census tract	CDC	Tract	Ex bet nei bu ac det
20	Lat/Lon	**SA Historic Districts** (COSA Planning)	data.sanantonio.gov	Spatial	Are ver sta
21	Year	**SA Major Events** — Fiesta SA, Spurs championships, NBA/NFL events	COSA / news archives	Year	Do co vol
22	ZIP / Tract	**ACS** — B25003 Tenure (Rent vs. Own)	Census	ZIP or tract	Re nei fac dis on

#	Primary Variable	External Dataset	Source	Merge Key	Po
23	Artist (Tejano/Mexican)	**Mexico Census / INEGI** — border region demographics	INEGI	Artist origin	Cro flov bric Me
24	Venue + Year	**Texas Historical Commission (THC)** — Recorded Texas Historic Landmarks	thc.texas.gov	Venue name / address	Ide pro cor sta
25	ZIP	**USPS ZIP Code Business Statistics**	Census ZIP Business Patterns	ZIP	Ver to t der mu ove in c
26	Year	**SA Convention & Visitors Bureau Annual Reports**	visitsanantonio.com	Year	Tou alo free cor driv
27	Lat/Lon	**SAMHD Alcohol Permits / TABC License Data**	TABC open data	Address	Co ver lice unl ver
28	Month + Venue	**TicketMaster/StubHub historical data** (via API or research datasets)	Proprietary / academic	Venue + Date	Act atte ava tra ana

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Section 3: Hypothetical Merges & Derived Insights (Detailed)

Merge A: Concert Geography × ACS Income & Race (ZIP/Tract Level)

How: Use `tidycensus::get_acs()` in R to pull B19013 (median HH income) and B02001 (race) at the ZIP Code Tabulation Area (ZCTA) level. Join to `Ven_Ev_Ar_Address` via the `Zip` field. For higher precision, use the lat/lon coordinates + `sf::st_join()` to join directly to

Census tract geometries (TIGER/Line shapefiles, `tigris::tracts(state="TX", county="Bexar")`).

Direct insights:

- Map concert density (events per ZIP over time) against median income — is live music a spatially privileged good in SA?
- Compute a "music access score" per neighborhood: events per capita, weighted by venue variety
- Test the journal's explicit hypothesis: southside vs. northside concert disparity by income and race

Indirect/creative insights:

- Build a **Cultural Redlining Index** — borrowing from housing redlining analysis — to show historical patterns of music venue exclusion in predominantly Hispanic south/west SA
- Compare the ZIP-level income gradient of venues over time: did the 1980s boom disproportionately benefit wealthy northside ZIPs?
- Use a **Dissimilarity Index** (standard segregation metric) adapted to measure how evenly concert access is distributed across racial groups in SA

Merge B: Artist Names × MusicBrainz Open Data

How: MusicBrainz provides a full Creative Commons database dump (XML or via their API) with artist hometown, genre, gender, formation/dissolution date. Fuzzy-match `artistsIndex.Name` against MusicBrainz `artist.name` using `fuzzyjoin` or `stringdist` in R.

Direct insights:

- What % of SA concert artists are local vs. touring?
- Did SA develop a local music ecosystem or rely primarily on national touring acts?
- Gender breakdown of performers across decades

Indirect/creative insights:

- Identify "SA-first" artists: acts who performed in SA before breaking nationally — did SA play a role in launching careers?
- Track artist "career arc" relative to SA performance frequency: do artists perform more in SA early, mid, or late career?

Merge C: Venue Geography × SA Neighborhood Change (Property Values / HUD FMR)

How: Use BCAD (Bexar County Appraisal District) parcel data (downloadable as CSV) matched by address to venue locations. HUD Fair Market Rent data is ZIP-level, annual, going back to 1983.

Direct insights:

- Which neighborhoods saw the largest property value increases and simultaneously lost venues?
- Quantify the economic displacement of music venues through gentrification

Indirect/creative insights:

- Build a **"venue survival curve"** (Kaplan-Meier-style) stratified by neighborhood income level at time of opening — do venues in low-income areas close faster?
- Test whether venue closures in a neighborhood preceded or followed rent increases (Granger causality-style temporal analysis)

Merge D: Concert Frequency × Recession/Economic Data (NBER + BLS)

How: NBER publishes official recession start/end dates. BLS provides SA-specific unemployment rates (BLS Local Area Unemployment Statistics, LAUS). Merge on `Year` (and `Month` for finer granularity).

Direct insights:

- Do concert counts drop during recessions? (Preliminary evidence: 1990–95 dip)
- Is live music a luxury good (income elasticity > 1) or a recession-resilient industry?

Indirect/creative insights:

- Compare SA's recession sensitivity to national concert industry trends (available from RIAA/NARM historical data)
- Identify whether specific genres (Country, Classical, Tejano) show different recession elasticity — genre as a proxy for audience economic class

Merge E: Genre Trends × Demographic Shifts (ACS Over Time)

How: ACS publishes estimates back to 2005; decennial Census provides 1970/1980/1990/2000/2010 breakdowns. Pair the decade-level genre data from the primary dataset with Census decade estimates of SA's Hispanic/Latino population share.

Direct insights:

- As SA's Hispanic population grew from ~50% (1970) to ~63% (2020), did Tejano/Latin genre representation in the dataset grow proportionally?
- If not, the gap quantifies collector bias with demographic backing

Indirect/creative insights:

- This is potentially publishable as a **music sociology paper**: "Demographic growth without cultural representation — the case of Tejano music in San Antonio concert records"
- Compare with similar analyses for Black music genres (R&B, Blues, Gospel) relative to SA's Black population trends

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Section 4: Multi-Variable Interaction Analysis

Table 2: Suggested Multi-Variable Interactions & Analytical Value

#	Variables Combined	Analytical Approach	Expected Finding / Value	Complexity
1	Year + Venue ZIP + Median Income	Time-series regression (lm / lme4)	Quantify income gradient of concert access over time; test if gap widened	Medium
2	Decade + Genre + % Hispanic (Census)	Correlation / chi-square	Test whether Tejano underrepresentation is statistically significant vs. demographic expectation	Medium
3	Lat/Lon + Venue lifespan + Property values	Survival analysis (Cox model)	Which neighborhood characteristics predict venue closure?	High
4	Month + Venue type + Season	ANOVA / Poisson regression	What drives seasonal concert patterns — weather, tourism, school calendar?	Low
5	Artist type (local/national) + Year + Decade	Longitudinal descriptive	Did SA's music scene become more or less locally driven over time?	Medium
6	Top venue events + ZIP income + Race	Geospatial clustering (k-means or DBSCAN)	Are high-activity venues clustered in high-income, low-diversity areas?	Medium

#	Variables Combined	Analytical Approach	Expected Finding / Value	Complexity
7	Genre + Day of week (derived) + Month	Heatmap / ANOVA	Do genre preferences vary by day or season? (e.g., Tejano on weekends, Classical on weekday evenings)	Low
8	Venue first/last year + Neighborhood gentrification index	Regression / event study	Causal test: does gentrification predict venue closure within 5 years?	High
9	Concert frequency/year + SA population growth	Per-capita normalization + trend	Is SA genuinely becoming a bigger music city, or just keeping pace with population?	Low
10	Artist name + MusicBrainz gender + Decade	Proportion test over time	Has gender diversity among performing artists improved decade over decade?	Medium

Recommended Statistical Models

Geospatial equity model (highest priority):

```
library(tidycensus); library(sf); library(spdep)
# Pull tract-level income + race
tracts <- get_acs("tract", variables = c("B19013_001","B03003_003"),
                  state="TX", county="Bexar", geometry=TRUE)
# Spatial join events to tracts
events_sf <- st_as_sf(ven_ev_ar_address, coords=c("Longitude","Latitude"),
crs=4326)
events_tracts <- st_join(events_sf, tracts)
# Model: events ~ income + pct_hispanic + decade (Poisson or NB)
library(MASS)
model <- glm.nb(n_events ~ median_income + pct_hispanic + decade,
data=tract_summary)
```

Venue survival analysis:

```
library(survival)
venue_lifespan <- events_sf %>%
  group_by(Venue) %>%
  summarise(start=min(Year), end=max(Year), lifespan=end-start) %>%
  left_join(tract_demographics, by="tract")
```

```
surv_fit <- survfit(Surv(lifespan) ~ income_quartile, data=venue_lifespan)
cox_model <- coxph(Surv(lifespan) ~ median_income + pct_hispanic +
decade_opened, data=venue_lifespan)
```

Genre-demographic correlation:

```
# Proportion test: expected vs. observed Tejano share
observed_tejano_pct <- sum(genres=="Tejano") / nrow(artist_genres_clean)
expected_pct <- 0.63  # SA Hispanic population share circa 2000
prop.test(x=tejano_count, n=total_events, p=expected_pct)
```

Section 5: Prioritization & Impact Assessment

Rank	Connection	Value	Access	Novelty	Justification
1	Events × ACS Income/Race by ZIP/Tract	Very High	Very Easy (`tidycensus`)	Medium	Directly tests the journal's own hypothesis; publishable; straightforward merge
2	Venue Geography × SA Neighborhood Map	Very High	Easy (COSA open data)	Medium	Enables all spatial analysis; prerequisite for most other merges
3	Genre × Hispanic Population Growth (Census)	Very High	Easy	High	Quantifies collector bias with demographic rigor; novel contribution
4	Artist × MusicBrainz (local vs. touring)	High	Medium (API rate limits)	High	Untouched in existing analysis; opens new research questions about SA's music ecosystem
5	Concert Frequency × Recession Dates (NBER/BLS)	High	Easy	Medium	Explains the 1990–95 dip with empirical backing

Rank	Connection	Value	Access	Novelty	Justification
6	Venue Lifespan × Property Values (BCAD/HUD)	High	Medium	Very High	Gentrification-music displacement is nationally relevant and underresearched in SA specifically
7	Events × Bexar County Population (Census)	Medium	Easy	Low	Necessary normalization step; low novelty on its own
8	Genre × Billboard Charts	Medium	Medium	High	Connects SA to national music industry trends
9	Venue × Business License / TABC Data	Medium	Medium	Medium	Operationalizes venue status (open/closed); enables displacement analysis
10	Month/Season × NOAA Weather	Low–Medium	Easy	Low	Useful descriptive context; limited analytical depth alone

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Section 6: Visualization & Reporting Recommendations

High-Priority Visualizations

Visualization	Variables	Tool	Purpose
Choropleth map — Event density by Census tract	Lat/Lon + ACS tract	`leaflet` + `sf` in R	Show spatial inequality in concert access
Choropleth map — Median income overlay on venue locations	ZIP/Tract income + venue lat/lon	`ggplot2` + `geom_sf`	Core equity finding visualization
Diverging bar chart — Tejano share vs. expected by Hispanic population %	Genre count + Census %	`ggplot2`	Visualize collector bias magnitude
Survival curve — Venue lifespan by neighborhood income quartile	Venue first/last year + income	`ggsurvfit`	Gentrification displacement evidence

Visualization	Variables	Tool	Purpose
Stacked area chart — Genre trends over decades	Decade + macro-genre	`ggplot2`	Requires macro-genre consolidation first
Scatter/bubble — Venue events vs. neighborhood income	Venue activity + ZIP income	`ggplot2` / `plotly`	Simple equity visual; accessible to non-technical audiences
Bivariate choropleth — Income × concert density simultaneously	ACS income + event count per tract	`biscale` R package	Sophisticated; shows areas that are low-income AND low-concert-access
Network graph — Artist co-appearance at venues	Artist × Venue co-occurrence	`igraph` / `ggraph`	Map SA's music community structure; who played together/at same venues
Recession overlay line chart — Concerts/year + recession shading	Year + NBER dates	`ggplot2` + `geom_rect`	Directly tests recession hypothesis from journal
Hexbin map — All geocoded events	Lat/Lon	`ggplot2` + `stat_binhex`	Density without overplotting; clear northside bias visualization

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Section 7: Gap Analysis

Gaps in Current Dataset

Gap	Impact	Recommended Data Source
68.8% of events lack coordinates	All spatial analysis is limited to 31% of geocoded data; northside bias in map may be artifact of geocoding gaps, not actual geographic distribution	Google Maps Geocoding API (batch geocode by address); OpenStreetMap Nominatim (free); Bexar CAD parcel data
No venue capacity data	Cannot compute events-per-capacity or saturation metrics	Setlist.fm API; manual research for top 20 venues; Texas TABC capacity records
No ticket price / attendance data	Cannot compute economic impact or affordability analysis	Historic newspaper archives (San Antonio Express-News digital archive); Pollstar

Gap	Impact	Recommended Data Source
		historical data (academic licensing)
No artist origin/hometown	Cannot distinguish local SA artists from touring acts	MusicBrainz API; AllMusic; Wikipedia API
Tejano / Spanish-language artist undercounting	Core research validity issue; conclusions about genre diversity are compromised	Tejano Music Awards archives (1981–present); SA Express-News music coverage; KXTN/KQQK historical playlists
`Event_Formulas` is 80% null	Cannot analyze event types (festivals, club nights, benefits, etc.)	Manual coding of top events; `stringr` pattern matching from `Event_Artists` raw field
Pre-1970 has no geographic data	Historical analysis cannot be spatially grounded	Sanborn Fire Insurance Maps (Library of Congress); SA Historic Map collection; newspaper archives
No demographic data on artists	Cannot analyze gender, race, or age representation	MusicBrainz; Wikidata SPARQL queries; manual coding for top 100 artists
No venue closure dates	Cannot distinguish active from closed venues	Bexar County business records; TABC; Google Places API historical data; SA Express-News
Genre micro-fragmentation (985 genres for 1,769 artists)	Makes aggregation and trend analysis unreliable	Build a macro-genre crosswalk table: map all 985 genres to ~10 parent categories using Spotify's genre taxonomy or manual coding

Recommended Immediate Actions

1. **Build a macro-genre crosswalk** — Map all 985 genres to ~10 parent categories (Country, Rock, Latin/Tejano, Classical, Blues/R&B, Jazz, Pop, Metal, Electronic, Other). This unlocks all genre trend analysis and is achievable in R with a lookup `case_when()` statement.
2. **Complete venue geocoding** — Use the Google Maps Geocoding API (free tier: 200 requests/month; or student research access) to geocode the 662 venues in `venues_with_coordinates` marked `needs_geo=TRUE`. This expands spatial coverage dramatically.
3. **Pull ACS data immediately** — `tidycensus` + ZIP/ZCTA join is executable in R in <20 lines. This delivers the single highest-value analysis (equity/access) with the least effort.

4. **Scrape Tejano Music Awards records** — A targeted web scrape of artist names from the TMA archives (1981–present) can serve as a ground-truth correction list for the Tejano genre undercounting.
5. **Create a `full_date` field** — Parse Month + Day + Year into a proper `Date` type in R using `lubridate::mdy(paste(Month, Day, Year))`. This enables time series modeling, day-of-week analysis, and proper gap analysis.

Summary: Top 5 Highest-Impact Next Steps

Priority	Action	Lines of R Code	Expected Deliverable
1	ACS income + race join via `tidycensus` (ZIP level)	~25	Choropleth map + equity summary table
2	Build macro-genre crosswalk	~30	Clean genre trend chart by decade
3	Geocode remaining 662 venues	API calls + join	Full spatial coverage for mapping
4	MusicBrainz artist metadata join	~50 (with fuzzyjoin)	Local vs. touring breakdown; gender analysis
5	Venue lifespan + HUD FMR overlay	~40	Gentrification-venue displacement analysis

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Report prepared for CEDISH "Sounds of San Anto" research project. All external data sources cited are publicly available or accessible via academic API access.